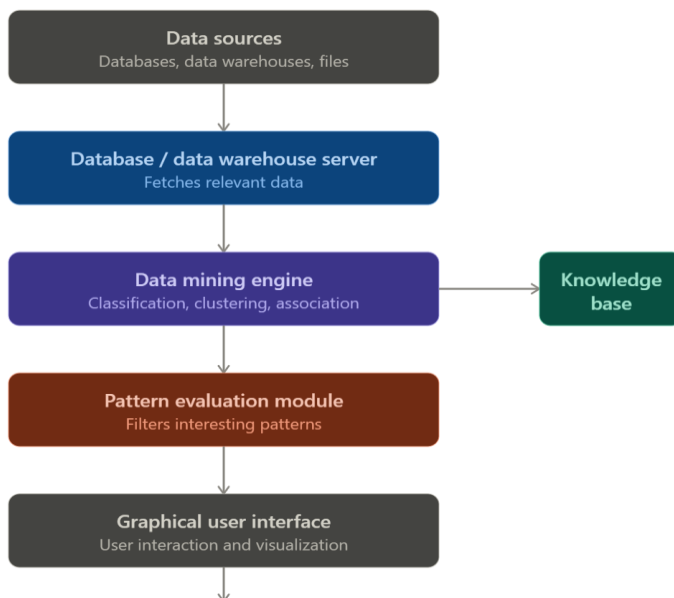


8 Marks

1. Four Components of Data Mining

1. **Data Source** – Databases, data warehouses, flat files, or the web that supply raw data for mining.
2. **Data Mining Engine** – Core component applying techniques like classification, clustering, and association rule mining.
3. **Pattern Evaluation Module** – Filters out uninteresting patterns using measures like support and confidence.
4. **User Interface/Knowledge Base** – Allows user interaction, query specification, and result visualization; knowledge base stores domain knowledge to guide mining.



2. Functionalities of Data Mining

1. **Characterization & Discrimination** – Summarizes/contrasts target class data.
2. **Association Rule Mining** – Finds co-occurring items (market basket analysis).
3. **Classification** – Predicts categorical labels using training data.
4. **Regression/Prediction** – Predicts continuous numeric values.
5. **Clustering** – Groups similar objects without predefined labels.
6. **Outlier Analysis** – Detects data that deviates from normal patterns.

3. Objective and Importance of Data Cleaning

Objective: To detect and correct/remove inaccurate, incomplete, duplicate, or inconsistent data so the dataset is reliable for mining (fills missing values, smooths noise, removes outliers, resolves inconsistencies).

Importance:

1. Improves data quality
2. Produces accurate mining results
3. Reduces redundancy
4. Handles missing data
5. Increases processing efficiency
6. Supports better decision-making

4. Explain Association Rule Mining

Definition: A technique to discover interesting relationships between items in transactional datasets, used in market basket analysis.

Rule form: $A \rightarrow B$ [support = s%, confidence = c%]

Key measures:

- $\text{Support}(A \rightarrow B) = P(A \cup B)$
- $\text{Confidence}(A \rightarrow B) = P(B|A)$

Example: {Bread, Butter} $\rightarrow$ {Milk}, support 20%, confidence 70%.

Algorithm: Apriori Algorithm — uses the property that any subset of a frequent itemset must also be frequent; generates candidate itemsets iteratively and prunes infrequent ones, then derives rules meeting minimum confidence.

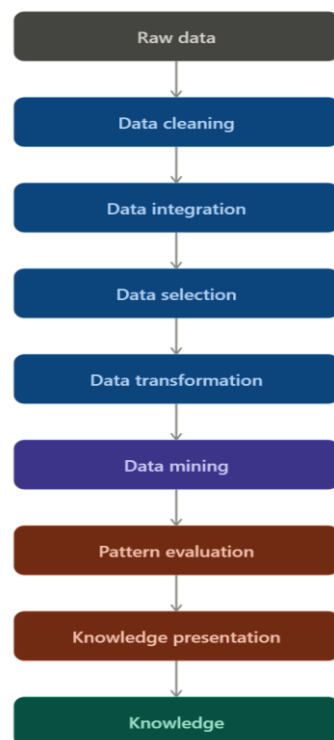
5. KDD Process

List the 5 core steps briefly (many 8-mark versions condense some steps):

1. **Data Cleaning** – Removes noise and handles missing values. It improves data quality.
2. **Data Integration** – Combines multiple data sources. Data from multiple sources is combined into a single dataset. Example: Combining customer information from sales and customer-service databases.
3. **Data Selection** – Retrieves relevant data for analysis. Relevant data is selected from one or more data sources. Only data required for the analysis is chosen.

4. **Data Transformation** – Converts data into forms suitable for mining (normalization, aggregation).
5. **Data Mining** – Applies algorithms to extract patterns. Techniques include classification, clustering, association, and prediction.
6. **Pattern Evaluation** – Identifies truly interesting patterns.
7. **Knowledge Presentation** - The final knowledge is presented to users. It may be displayed using graphs, charts, tables, reports, or visualization tools.

Simple KDD Flow



6. Applications, Advantages, and Disadvantages of Data Mining

Applications :

- Market basket analysis
- Fraud detection
- CRM/customer segmentation
- Healthcare diagnosis
- Telecommunication
- Web mining/recommendation systems.

Advantages:

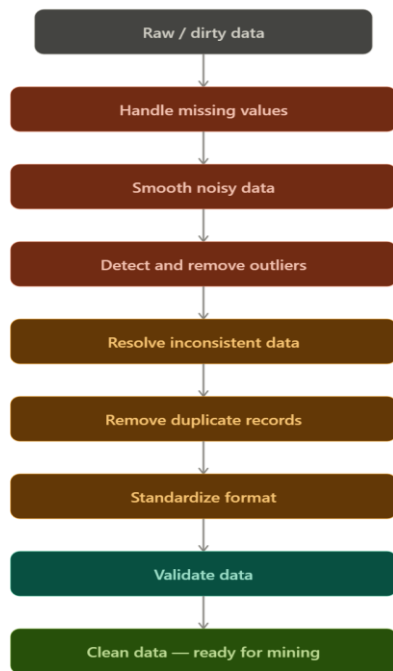
1. Improved decision-making
2. Detects hidden patterns
3. Improves customer targeting/satisfaction
4. Increases revenue via cross-selling
5. Fraud Detection.
6. Risk Management

Disadvantages:

1. Privacy concerns
2. High implementation cost
3. Requires technical expertise
4. Risk of inaccurate results from poor-quality data
5. Misuse of Information
6. Security Risk

7. Complete Data Cleaning Process

1. **Handling Missing Values** – Ignore tuple, fill manually, use global constant/mean, or predict using regression.
2. **Handling Noisy Data** – Techniques: Binning, Regression, Clustering.
3. **Outlier Detection** – Identify and remove/adjust extreme deviating values.
4. **Resolving Inconsistent Data** – Correct naming/format conflicts using metadata rules.
5. **Removing Duplicates** – Detect and eliminate duplicate records from merged sources.



10 Marks

1) Four Components of Data Mining.

1. **Data Source** – Includes structured/unstructured data from multiple systems; must be cleaned and integrated before mining.
2. **Database or Data Warehouse Server** – Fetches relevant data based on the user's mining request; acts as a bridge between raw storage and the mining engine.
3. **Data Mining Engine** – Contains a set of functional modules for tasks such as characterization, association, classification, clustering, and outlier analysis.
4. **Pattern Evaluation Module** – Works closely with the mining engine, using interestingness thresholds (support, confidence, lift) to focus the search toward interesting patterns.
5. **Graphical User Interface (GUI)** – Enables users to interact with the system, browse results, visualize patterns in charts/tables, and specify mining requests.
6. **Knowledge Base** – Domain knowledge that helps guide the search or evaluate discovered patterns' interestingness (e.g., concept hierarchies).

2) Functionalities of data mining

1. **Characterization** – Summarizes general features of a target class, e.g., characterizing high-spending customers by age, income. **Discrimination** compares this class against a contrasting class (e.g., low-spending customers).

2. **Association Rule Mining** – Discovers rules like {Bread, Butter}→{Milk}; used in retail cross-selling strategies.
3. **Classification** – E.g., classifying bank loan applicants as "approved" or "rejected" using decision trees or Naïve Bayes.
4. **Regression/Prediction** – E.g., predicting a house's sale price based on area, location, and amenities.
5. **Clustering** – E.g., segmenting customers into groups for targeted marketing using k-means.
6. **Outlier/Anomaly Detection** – E.g., detecting fraudulent credit card transactions that differ from a customer's usual spending pattern.
7. **Evolution/Trend Analysis** – Studies time-based data to identify trends and regularities, e.g., analyzing stock price movement or seasonal sales trends over years.

3) Objective and importance of data cleaning

Objective:

The objective of data cleaning is to ensure the dataset used for KDD/mining is complete, accurate, consistent, and free from noise/duplication. This involves filling in missing values, identifying and smoothing noisy data, detecting and removing outliers, correcting inconsistent formats/labels, and eliminating duplicate records — so the "garbage in, garbage out" problem is avoided.

Importance (expanded with explanation for each):

1. **Improves Data Quality** – Ensures data truly reflects real-world entities and relationships.
2. **Better Mining Results** – Clean input data produces more accurate patterns, classifications, and predictions.
3. **Reduces Redundancy** – Eliminating duplicates saves storage space and prevents skewed statistical results.
4. **Handles Missing Data** – Prevents bias or errors that would arise from incomplete records being used directly in models.
5. **Increases Efficiency** – Reduces processing time and computational cost of subsequent mining algorithms.
6. **Ensures Consistency** – Resolves conflicts that occur when integrating data from multiple heterogeneous sources (e.g., different naming conventions).
7. **Supports Reliable Decision-Making** – Business and scientific decisions based on clean data are more trustworthy and defensible.

8. **Prevents Costly Errors** – In domains like healthcare/finance, dirty data can lead to serious real-world consequences (e.g., wrong diagnosis, wrong credit decision).

4) Explain association rule mining

Definition: Association rule mining discovers interesting correlations, frequent patterns, or associations among sets of items in large transactional or relational databases — most famously used in market basket analysis.

Rule form: $A \rightarrow B$ [support = s%, confidence = c%], meaning if a transaction contains itemset A, it's likely to also contain itemset B.

Measures:

- **Support($A \rightarrow B$)** = (Transactions containing $A \cup B$) / (Total transactions) — measures frequency/popularity.
- **Confidence($A \rightarrow B$)** = (Transactions containing $A \cup B$) / (Transactions containing A) — measures reliability/strength of the rule.
- **Lift($A \rightarrow B$)** = Confidence($A \rightarrow B$) / Support(B) — measures how much more likely B is purchased when A is purchased compared to random chance (Lift > 1 = positive correlation).

Apriori Algorithm Steps (detailed):

1. Scan the database to find support count of each single item; retain those meeting minimum support (L1).
2. Generate candidate itemsets of size 2 (C2) by joining L1 with itself; scan DB, retain frequent ones (L2).
3. Repeat: generate C_k from $L_{(k-1)}$ using the join step, apply the **Apriori pruning property** ("any subset of a frequent itemset must be frequent") to eliminate candidates with infrequent subsets.
4. Continue until no more frequent itemsets can be generated.
5. From all frequent itemsets, generate association rules that satisfy minimum confidence threshold.

Example: In a supermarket dataset, if $\{\text{Bread, Butter}\} \rightarrow \{\text{Milk}\}$ has support = 20% and confidence = 70%, it means 20% of all transactions contain all three items, and 70% of customers buying bread and butter also buy milk.

Applications:

- Market basket analysis
- Fraud detection
- CRM/customer segmentation

- Healthcare diagnosis
- Telecommunication
- Web mining/recommendation systems.

5) KDD process

KDD is a **non-trivial, iterative, and interactive process**.

1. **Data Cleaning** – Removes noise, corrects inconsistent data, and fills/handles missing values to improve data quality.
2. **Data Integration** – Merges data from multiple heterogeneous sources (databases, files, flat tables) into a single coherent data store, resolving schema conflicts.
3. **Data Selection** – Relevant data for the specific analysis task is retrieved/extracted from the integrated data store.
4. **Data Transformation** – Data is transformed into forms appropriate for mining through normalization, aggregation, generalization, or attribute construction.
5. **Data Mining** – The essential step where intelligent methods (classification, clustering, association rule mining, regression, etc.) are applied to extract hidden patterns.
6. **Pattern Evaluation** – Identifies the genuinely interesting/useful patterns representing knowledge based on interestingness measures (support, confidence, novelty).
7. **Knowledge Presentation** – Visualization techniques (graphs, tables, reports, decision trees) present the mined knowledge to end users in an understandable form.

6) Applications, advantages and disadvantage of data mining

Applications:

1. Market Basket Analysis (retail cross-selling)
2. Fraud Detection (banking/credit cards)
3. Customer Relationship Management (targeted marketing)
4. Healthcare/Medical Diagnosis (disease prediction)
5. Education (predicting student performance/dropouts)
6. Telecommunications (churn prediction)
7. Financial Analysis (stock prediction, risk analysis, loan approval)
8. Web Mining (search engines, recommendation systems like Netflix/YouTube)

9. Intrusion Detection/Cybersecurity (anomaly detection in network traffic)

Advantages:

1. Enables data-driven, informed business decisions
2. Detects hidden trends/patterns not visible via simple queries
3. Improves customer satisfaction through personalization
4. Increases revenue through targeted marketing and cross-selling
5. Automates large-scale pattern discovery, saving manual effort
6. Improves security via fraud/anomaly detection
7. Supports predictive analytics for forecasting future outcomes

Disadvantages:

1. **Privacy Concerns** – Mining personal data can violate user privacy.
2. **Security Risks** – Sensitive data may be misused if not properly protected.
3. **High Cost** – Requires expensive tools, infrastructure, and skilled personnel.
4. **Data Quality Dependency** – Poor input data produces inaccurate/misleading patterns.
5. **Complexity** – Requires technical expertise to build models and interpret results correctly.
6. **Misuse of Information** – Discovered patterns can be misused for discriminatory practices (e.g., biased lending or insurance decisions).
7. **Accuracy Issues** – Overfitting or wrong algorithm choice can lead to false or misleading patterns.

7) Explain complete Data cleaning process.

1. Handling Missing Values

- Ignore the tuple (used when class label is missing, effective only for large datasets)
- Fill manually (accurate but time-consuming, infeasible for large data)
- Use a global constant (e.g., "Unknown")
- Use attribute mean/median (overall or class-wise)
- Use the most probable value via regression, Bayesian inference, or decision tree induction

2. Handling Noisy Data

- **Binning** – Sort data into bins, then smooth using bin mean, median, or boundaries
- **Regression** – Fit data into a regression function to smooth values
- **Clustering** – Group similar values; points outside clusters flagged as outliers
- **Combined computer and human inspection** – Computer flags suspicious values; humans verify manually

3. Identifying and Removing Outliers

Detected using statistical tests (z-score), clustering, or visualization (box plots); removed or treated separately since they distort analysis results.

4. Resolving Inconsistent Data

Correcting inconsistencies in naming/coding conventions (e.g., "M/F" vs. "Male/Female") using knowledge engineering tools, metadata, and data auditing/scrubbing tools.

5. Removing Duplicate Data (Deduplication)

Identifying duplicate records — often arising from merging multiple sources — using matching keys or similarity measures, and eliminating them to avoid biased/skewed results.

6. Data Standardization/Formatting

Ensures uniform formats across the dataset — standardizing date formats, currency units, and measurement units.

7. Validation

Post-cleaning, data is validated against defined constraints (e.g., age cannot be negative) to confirm integrity before moving to transformation/mining.

Summary Flow:

Raw Data → Detect & Handle Missing Values → Detect & Smooth Noise/Outliers → Resolve Inconsistencies → Remove Duplicates → Standardize Format → Validate → Clean Data ready for mining.